

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of seed mass compared to reference sites

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’	

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
```

```
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_refs.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = col_factor(levels = c("moist", "fresh", "dry"), ordered = TRUE),
    mngm.type = "f"
  )
) %>%
  rename(y = cwm.abu.seedmass)
```

Load data

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.001572228 0.001487247 0.001402265
```

```
median(sites$y)
```

```
## [1] 0.00121
```

```
sd(sites$y)
```

```
## [1] 0.001087929
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
## 0.000370 0.003408
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct> <int>
## 1 654      209
## 2 664      212
## 3 686      211
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord> <int>
## 1 positive      57
## 2 restored     472
## 3 negative     103
```

```
sites %>% count(hydrology)
```

```
## # A tibble: 3 x 2
##   hydrology      n
##   <ord> <int>
## 1 moist     135
## 2 fresh     327
## 3 dry       170
```

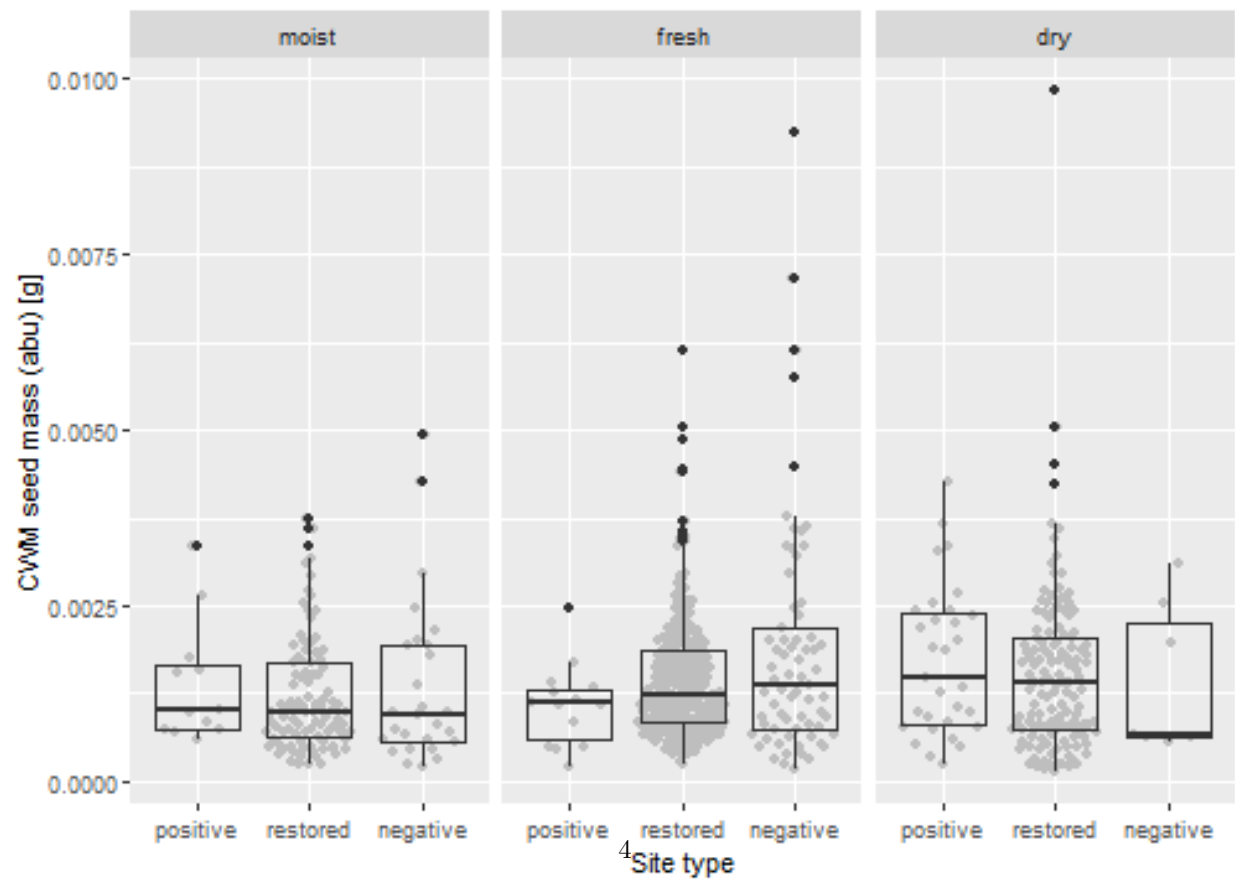
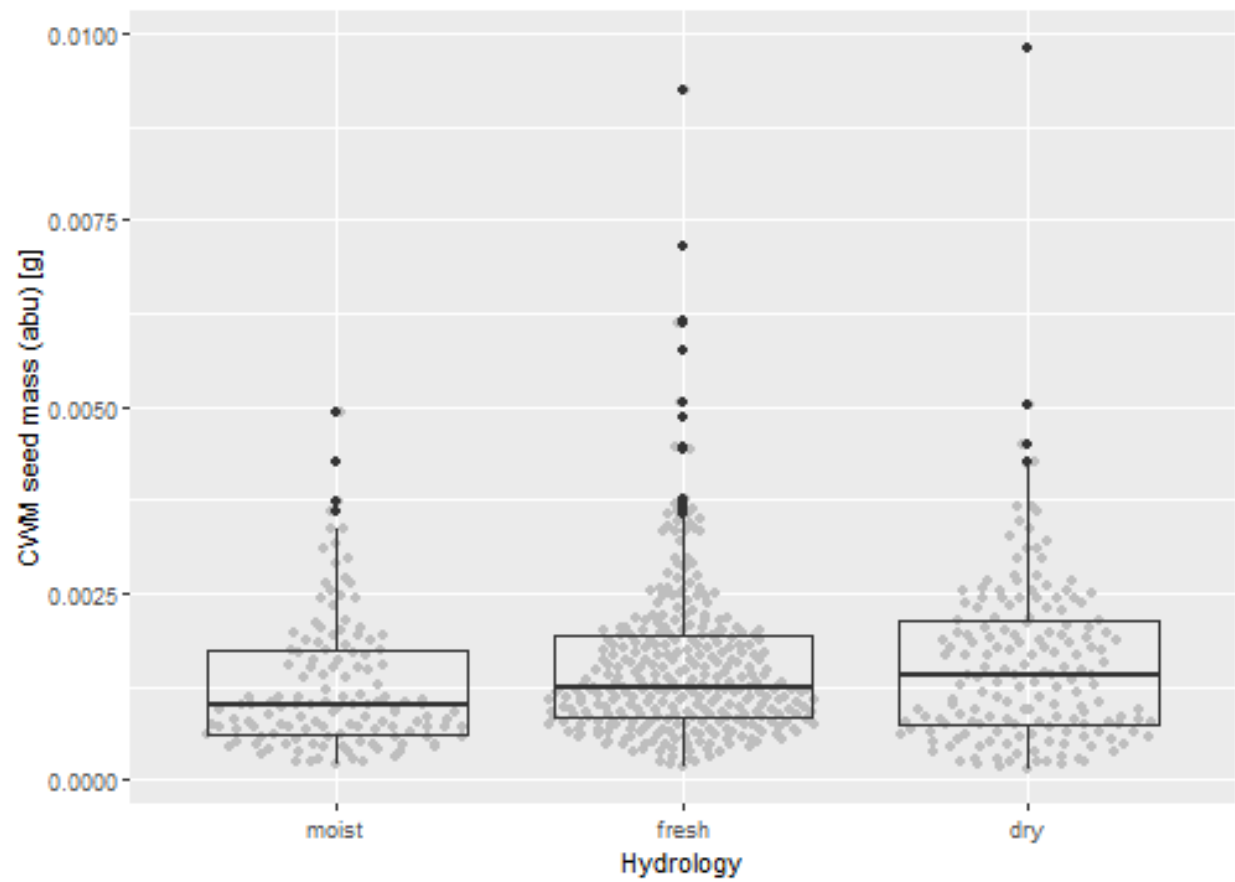
```
sites %>% count(hydrology, eco.id)
```

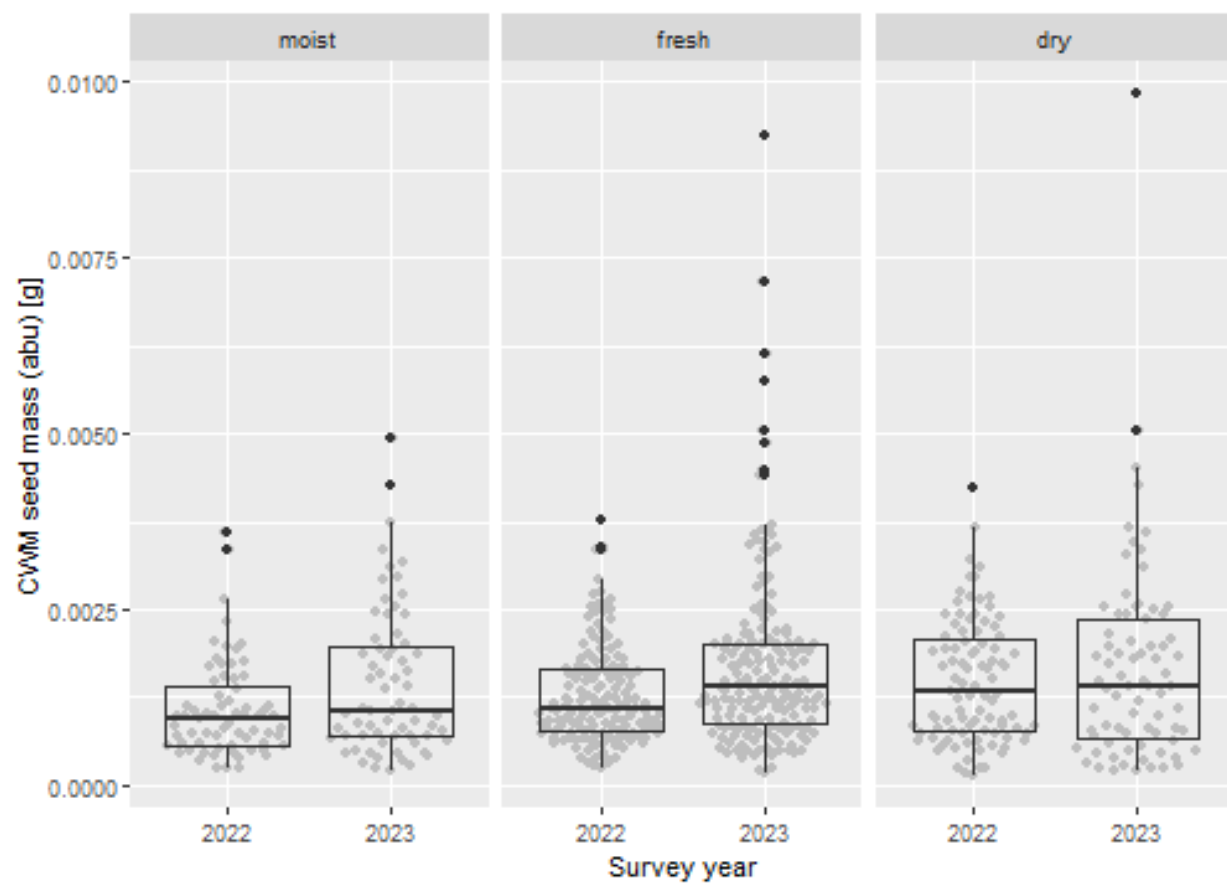
```
## # A tibble: 9 x 3
##   hydrology eco.id      n
##   <ord> <fct> <int>
## 1 moist   654      35
## 2 moist   664      50
## 3 moist   686      50
## 4 fresh   654     108
## 5 fresh   664     109
## 6 fresh   686     110
## 7 dry     654      66
## 8 dry     664      53
## 9 dry     686      51
```

```
sites %>% count(hydrology, site.type)
```

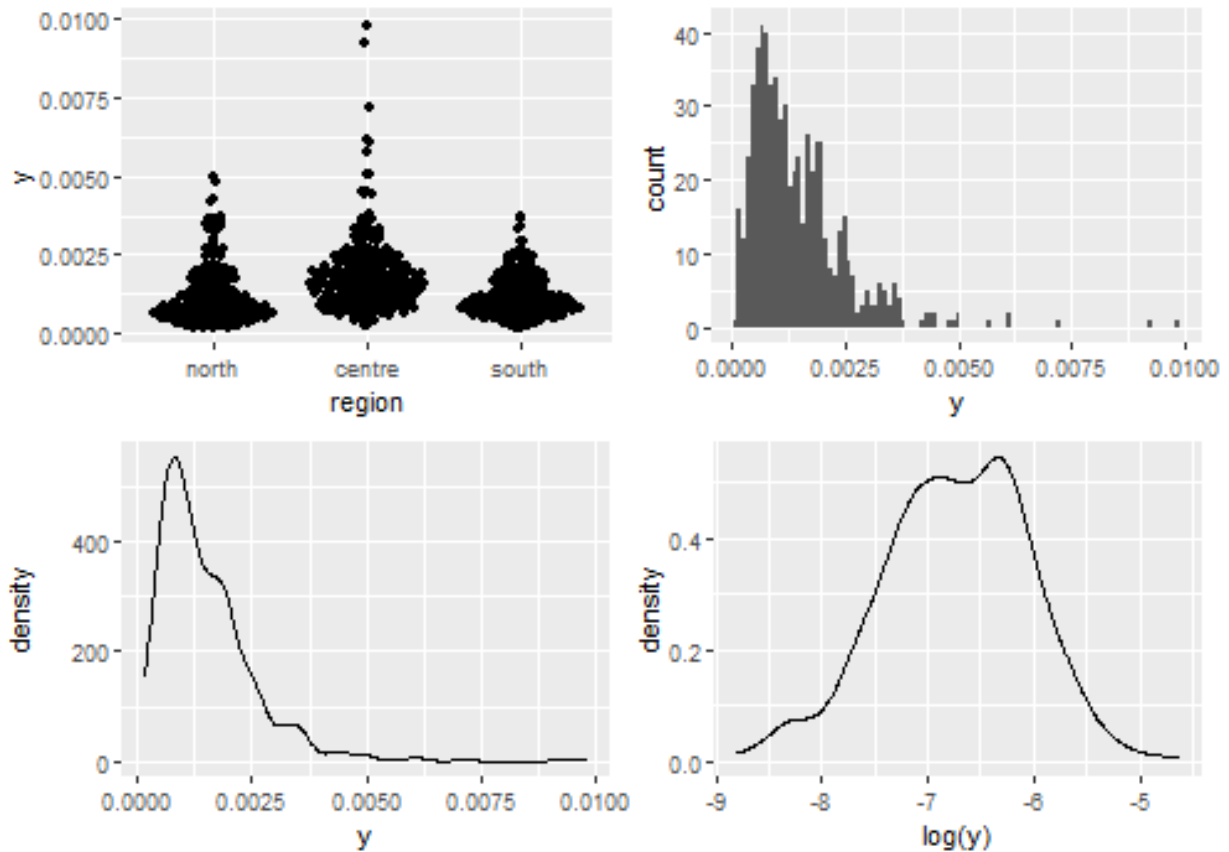
```
## # A tibble: 9 x 3
##   hydrology site.type      n
##   <ord> <ord> <int>
## 1 moist   positive     12
## 2 moist   restored     96
## 3 moist   negative     27
## 4 fresh   positive     14
## 5 fresh   restored    244
## 6 fresh   negative     69
## 7 dry     positive     31
## 8 dry     restored    132
## 9 dry     negative      7
```

Graphs of raw data (Step 2, 6, 7)





Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 Ecology DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_seedmass_refs_1.Rdata"))
load(file = here("outputs", "models", "model_seedmass_refs_2.Rdata"))
m_1 <- m1
m_2 <- m2

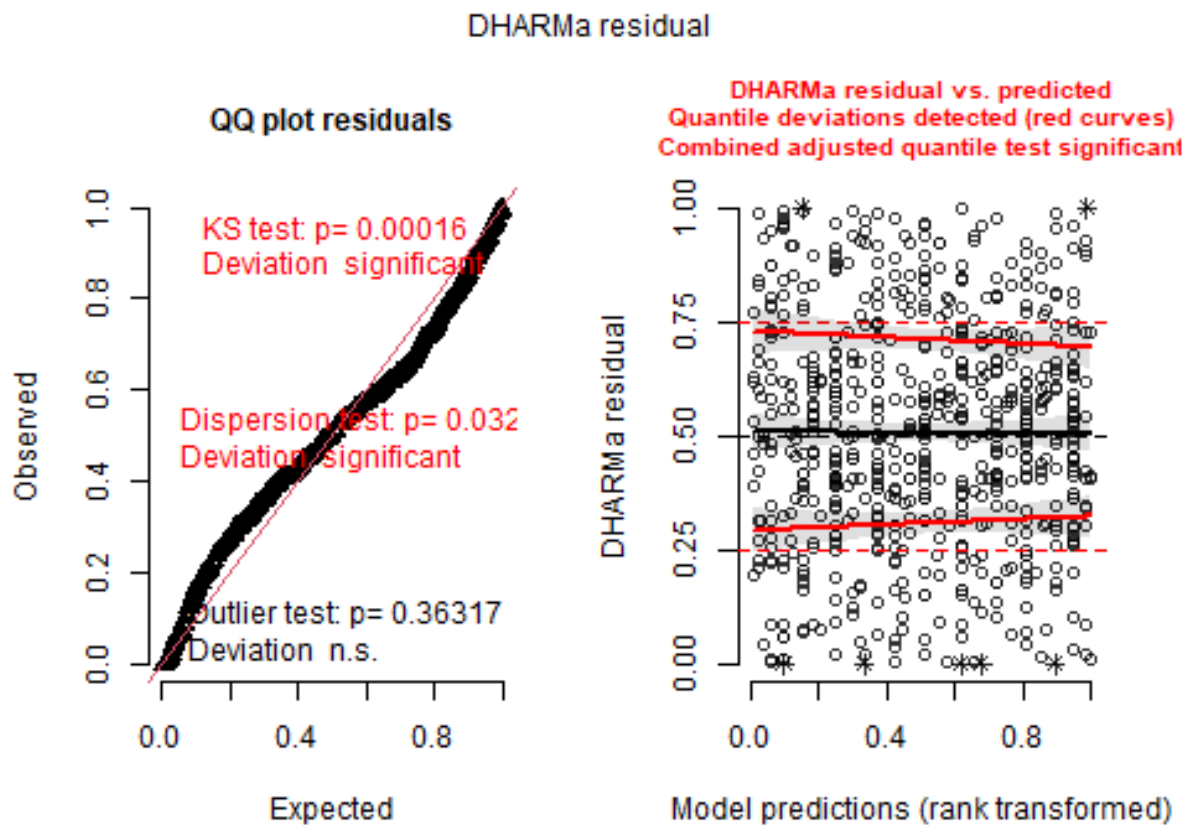
m_1@call
## lmer(formula = log(y) ~ hydrology * site.type + eco.id + obs.year +
```

```
##      (1 | id.site), data = sites, REML = FALSE)
m_2@call
## lmer(formula = log(y) ~ (hydrology + site.type + eco.id + obs.year)^2 +
##      (1 | id.site), data = sites, REML = FALSE)
```

Model check

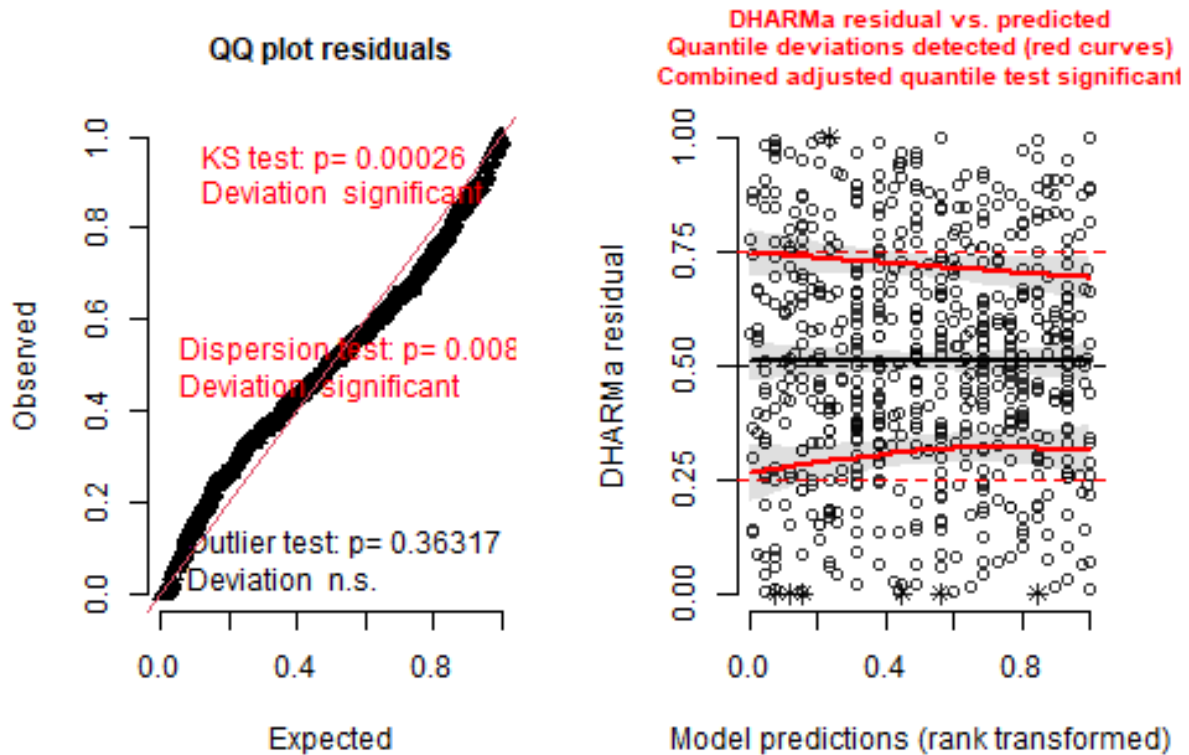
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

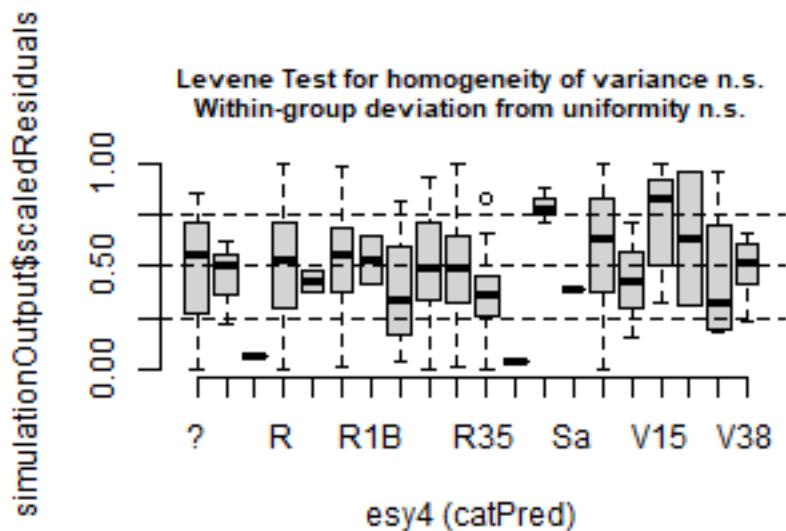


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

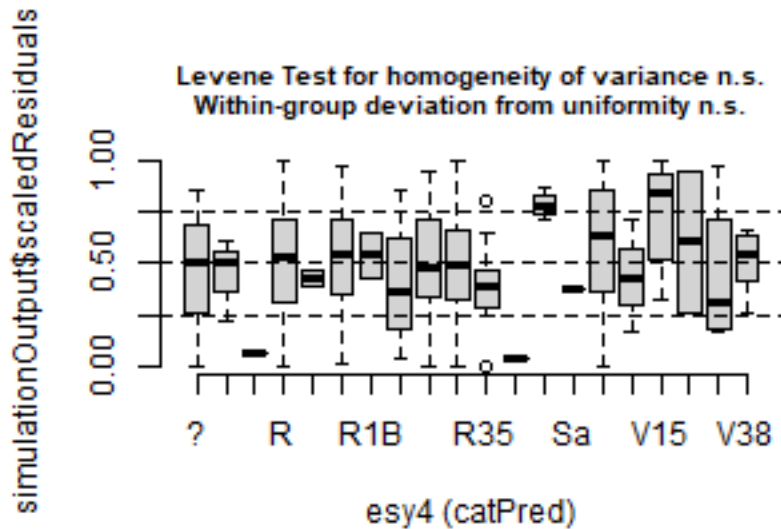
DHARMa residual



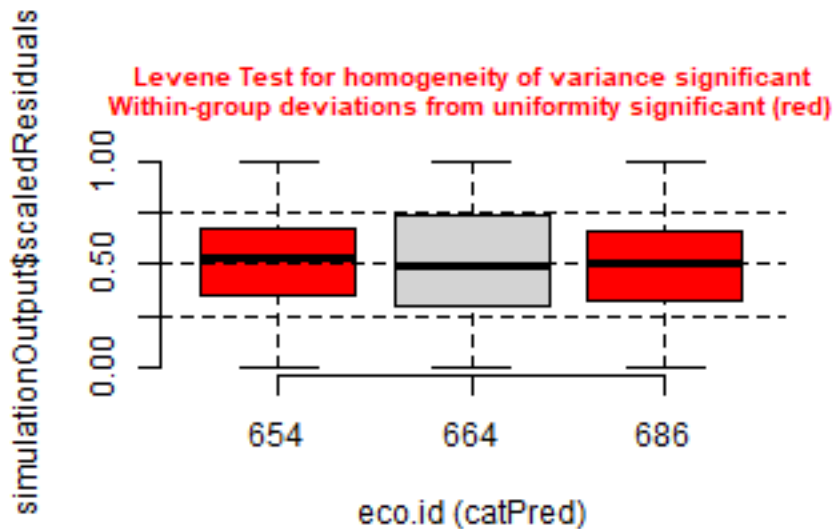
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMa::ensurePredictor: character string was provided as predictor. DHARMa
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMa.
```



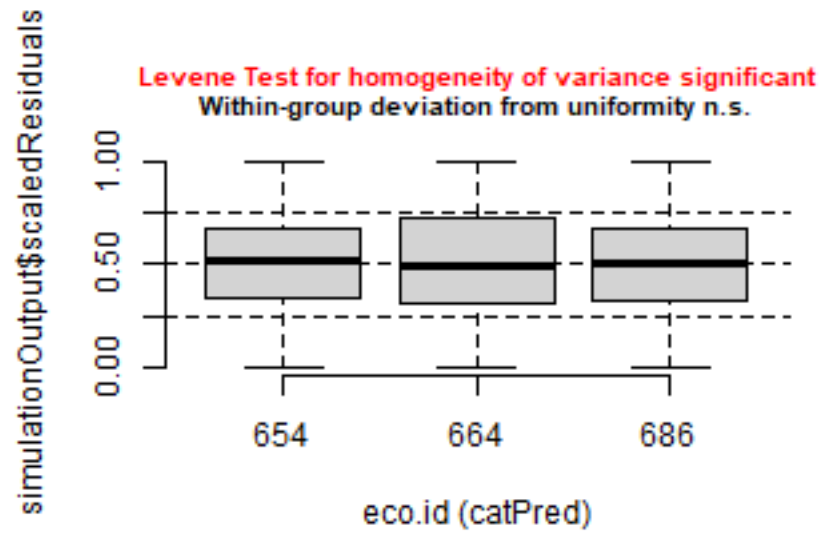
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
## Warning in ensurePredictor(simulationOutput, predictor):
## DHARMa::ensurePredictor: character string was provided as predictor. DHARMa
## has converted to factor automatically. To remove this warning, please convert
## to factor before attempting to plot with DHARMa.
```



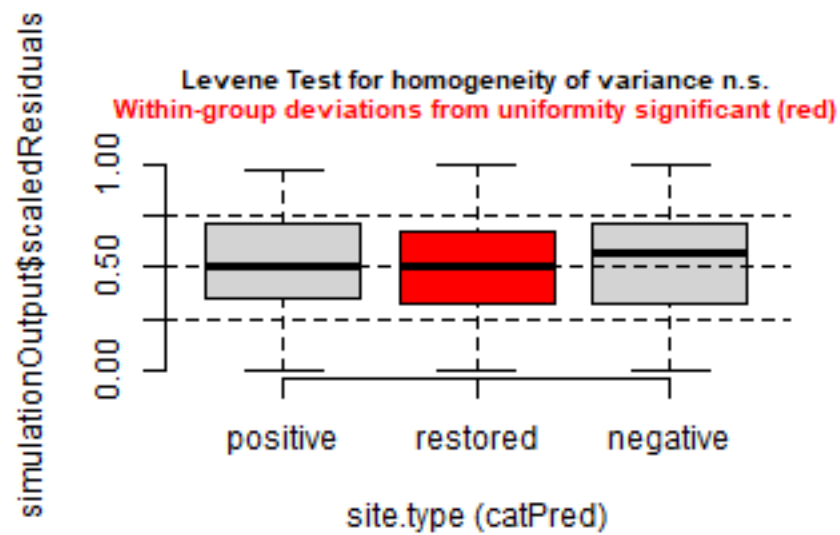
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



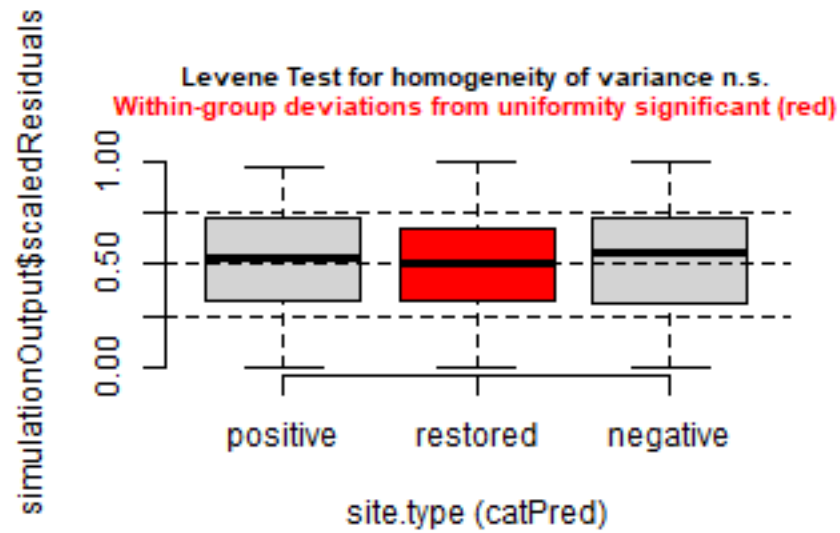
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



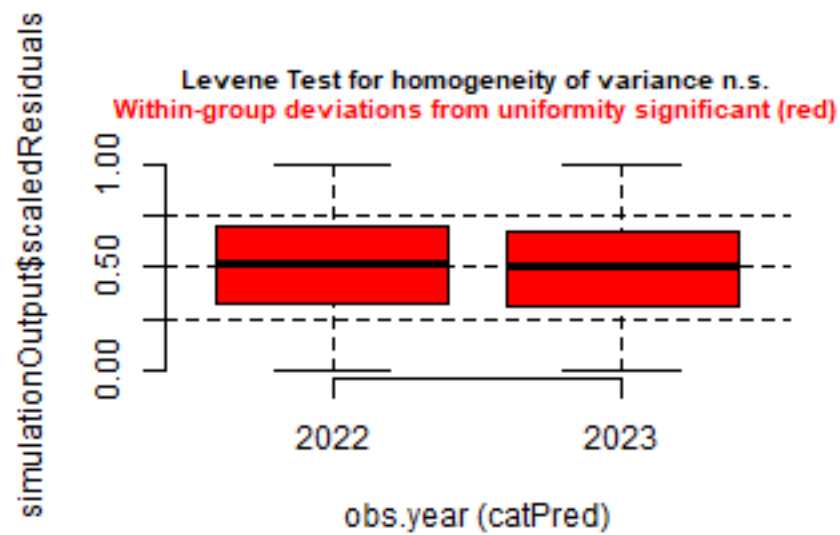
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



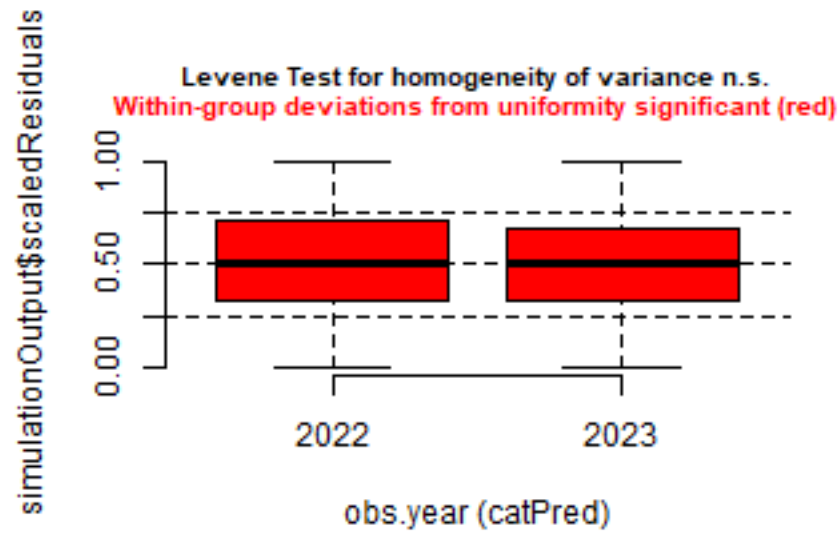
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



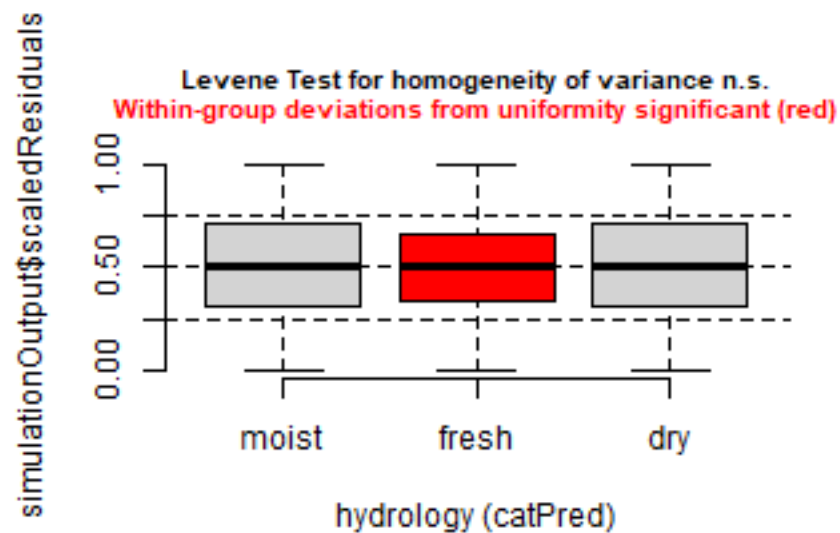
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



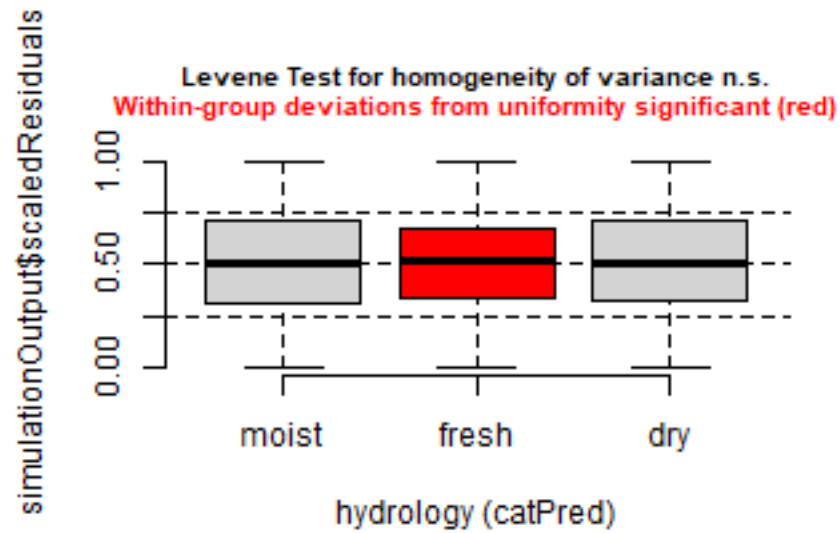
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



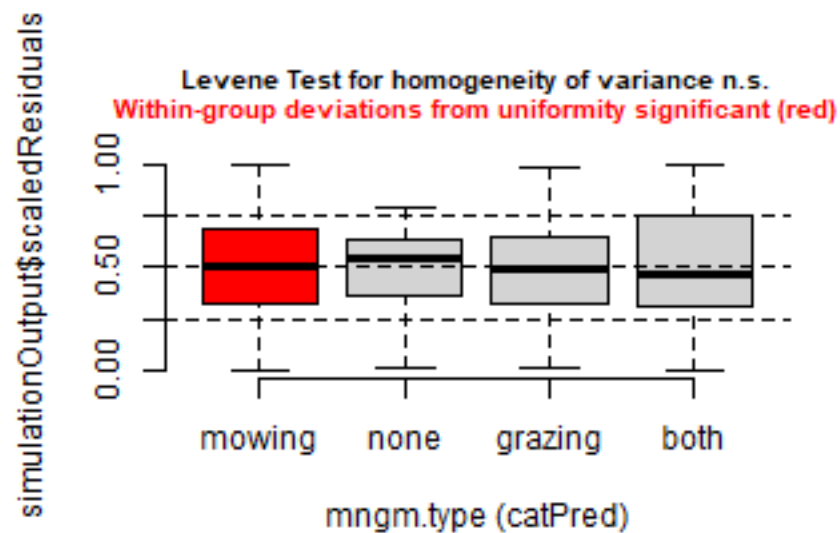
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



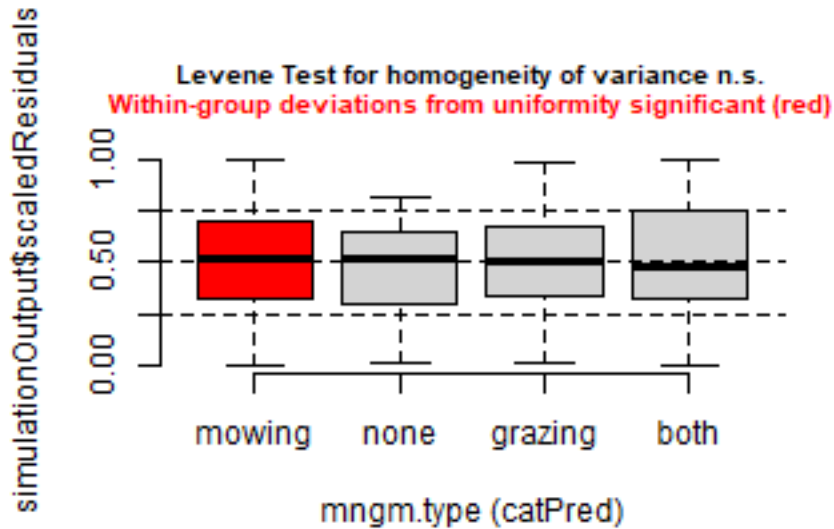
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
```

```
## method from
```

```
## na.action.merMod lme4
```

```
##          GVIF Df GVIF^(1/(2*Df))
## hydrology      5.148144 2      1.506304
## site.type      2.213162 2      1.219701
## eco.id          1.038971 2      1.009604
## obs.year        1.059859 1      1.029495
## hydrology:site.type 7.097400 4      1.277578
```

```
car::vif(m_2)
```

```
##          GVIF Df GVIF^(1/(2*Df))
## hydrology      44.713534 2      2.585888
## site.type      30.625977 2      2.352461
## eco.id          9.821478 2      1.770289
## obs.year        4.389900 1      2.095209
## hydrology:site.type 14.582019 4      1.397904
## hydrology:eco.id  15.875451 4      1.412833
## hydrology:obs.year  5.506121 2      1.531833
## site.type:eco.id  29.876378 4      1.529030
## site.type:obs.year 11.382068 2      1.836772
## eco.id:obs.year    9.086215 2      1.736184
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m           R2c
## [1,] 0.1407488 0.654619
MuMIn::r.squaredGLMM(m_2)
##           R2m           R2c
## [1,] 0.2679103 0.6533638
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_2 28 987.6853
## m_1 14 999.1688
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: log(y)
##               Chisq Df Pr(>Chisq)
## (Intercept)    4295.5618  1 < 2.2e-16 ***
## hydrology         0.3523  2   0.83848
## site.type         0.0200  2   0.99004
## eco.id            27.1322  2  1.283e-06 ***
## obs.year          4.3891  1   0.03617 *
## hydrology:site.type 3.8233  4   0.43045
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: log(y) ~ hydrology * site.type + eco.id + obs.year + (1 | id.site)
## Data: sites
##
##           AIC           BIC      logLik -2*log(L)  df.resid
##      998.5       1060.8      -485.2      970.5       618
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.7555 -0.4560  0.0106  0.4787  2.7887
##
## Random effects:
## Groups Name Variance Std.Dev.
## id.site (Intercept) 0.2450  0.4950
## Residual          0.1647  0.4058
## Number of obs: 632, groups: id.site, 172
```

```
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)    -6.561066   0.100107 -65.541
## hydrology.L      0.071631   0.126502   0.566
## hydrology.Q     -0.006139   0.103754  -0.059
## site.type.L      0.018310   0.135548   0.135
## site.type.Q      0.001527   0.089321   0.017
## eco.id664       -0.490866   0.102423  -4.793
## eco.id686       -0.424990   0.101902  -4.171
## obs.year2023     0.178139   0.085030   2.095
## hydrology.L:site.type.L -0.044833   0.257999  -0.174
## hydrology.Q:site.type.L  0.344000   0.210372   1.635
## hydrology.L:site.type.Q -0.012183   0.170754  -0.071
## hydrology.Q:site.type.Q  0.159146   0.139646   1.140
##
## Correlation of Fixed Effects:
##      (Intr) hydr.L hydr.Q st.t.L st.t.Q ec.664 ec.686 o.2023 h.L:...L
## hydrology.L  0.144
## hydrology.Q  0.121  0.203
## site.type.L -0.077 -0.425 -0.398
## site.type.Q  0.485  0.275  0.219 -0.180
## eco.id664   -0.496  0.096 -0.011 -0.072  0.021
## eco.id686   -0.544  0.052  0.038 -0.045 -0.011  0.502
## obs.yer2023 -0.460 -0.034  0.185 -0.035 -0.040 -0.004  0.030
## hydr1.L:...L -0.289 -0.371 -0.327  0.302 -0.383 -0.017  0.059 -0.014
## hydr1.Q:...L -0.297 -0.340  0.087  0.326 -0.366 -0.067  0.028  0.100  0.243
## hydr1.L:...Q  0.206  0.709  0.197 -0.380  0.201  0.042 -0.020 -0.070 -0.340
## hydr1.Q:...Q  0.060  0.206  0.727 -0.362  0.321 -0.018  0.049  0.173 -0.296
##      h.Q:...L h.L:...Q
## hydrology.L
## hydrology.Q
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## hydr1.L:...L
## hydr1.Q:...L
## hydr1.L:...Q -0.313
## hydr1.Q:...Q  0.081  0.138
```

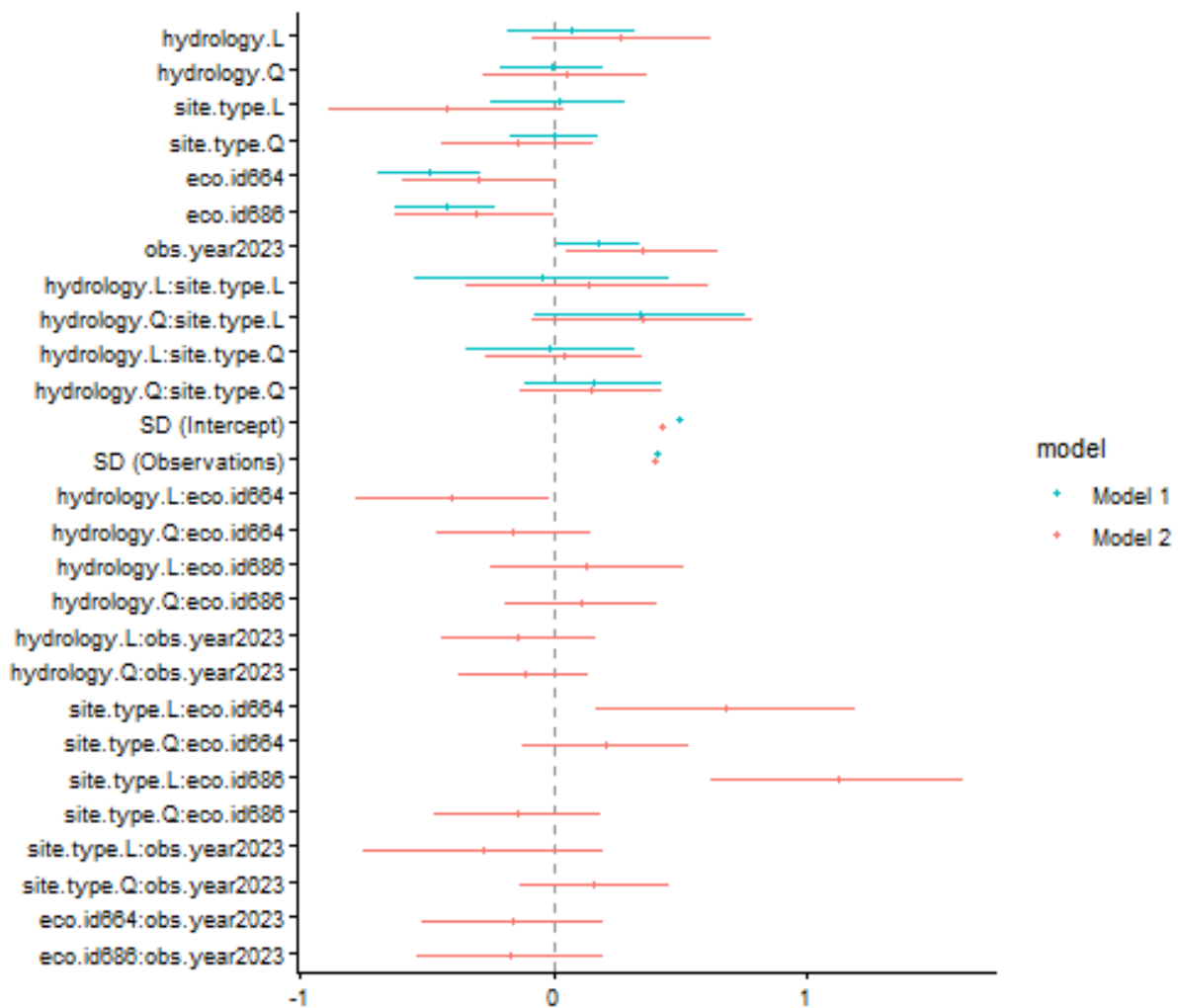
Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
```

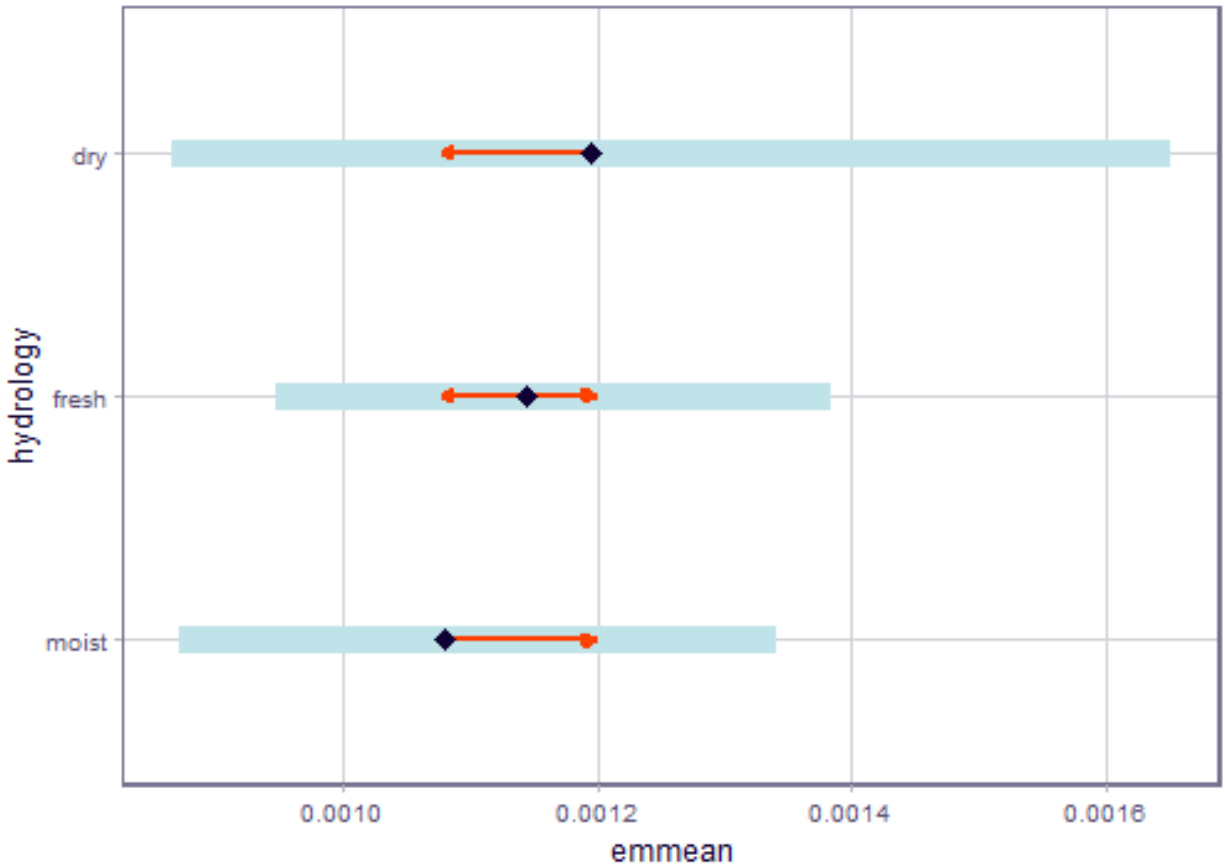
```
## for random effect parameters.

## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
## Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes: Hydrology

```
emm <- emmeans(
  m_1,
  revpairwise ~ hydrology,
  type = "response"
)
plot(emm, comparison = TRUE)
```

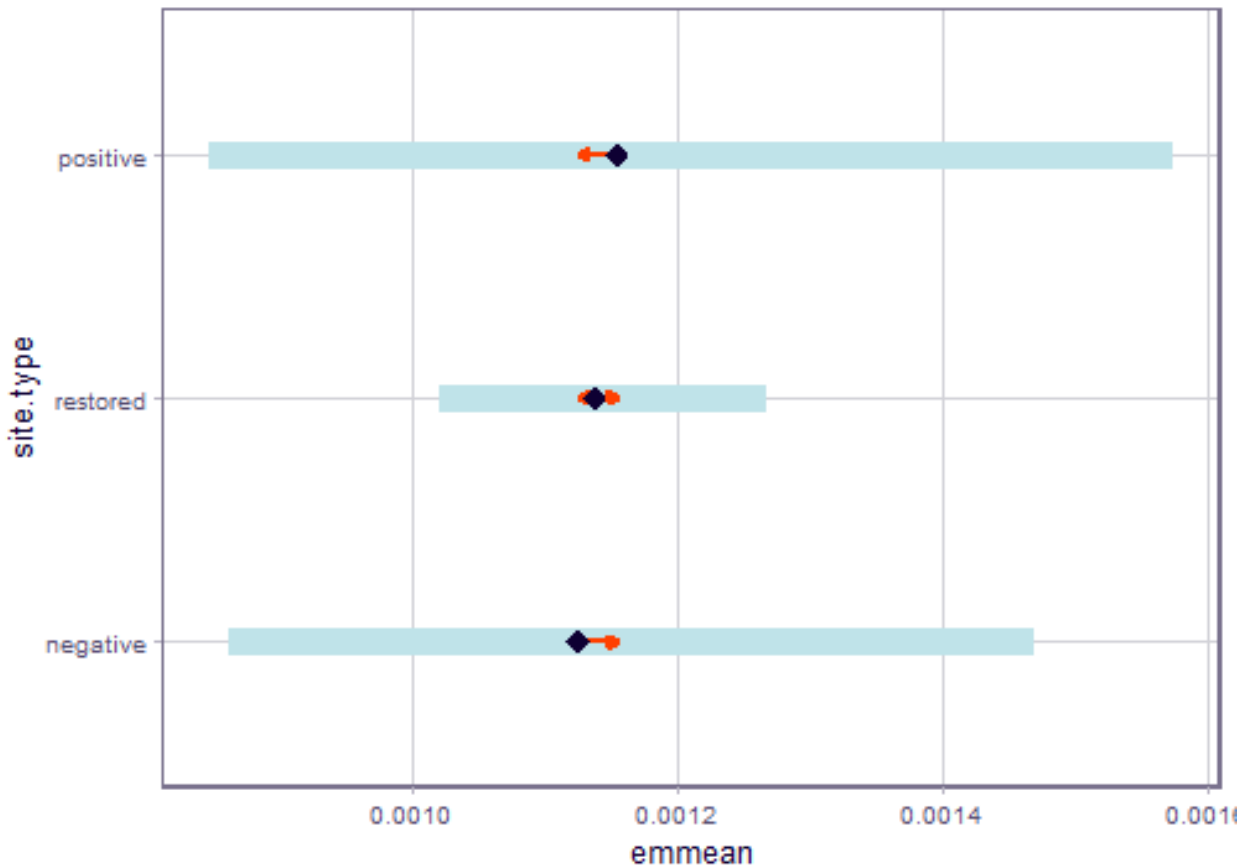


```
emm
```

```
## $emmeans
## hydrology response      SE    df lower.CL upper.CL
## moist      0.00108 0.000118 167.7 0.000871 0.00134
## fresh      0.00115 0.000106  26.8 0.000947 0.00138
## dry        0.00120 0.000178  12.8 0.000866 0.00165
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## contrast      ratio    SE   df null t.ratio p.value
## fresh / moist  1.06 0.153 61.9   1   0.403 0.9145
## dry / moist    1.11 0.205 22.8   1   0.547 0.8489
## dry / fresh    1.04 0.183 15.6   1   0.246 0.9673
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
```

Effect sizes: Site type

```
emm <- emmeans(
  m_1,
  revpairwise ~ site.type,
  type = "response"
)
plot(emm, comparison = TRUE)
```



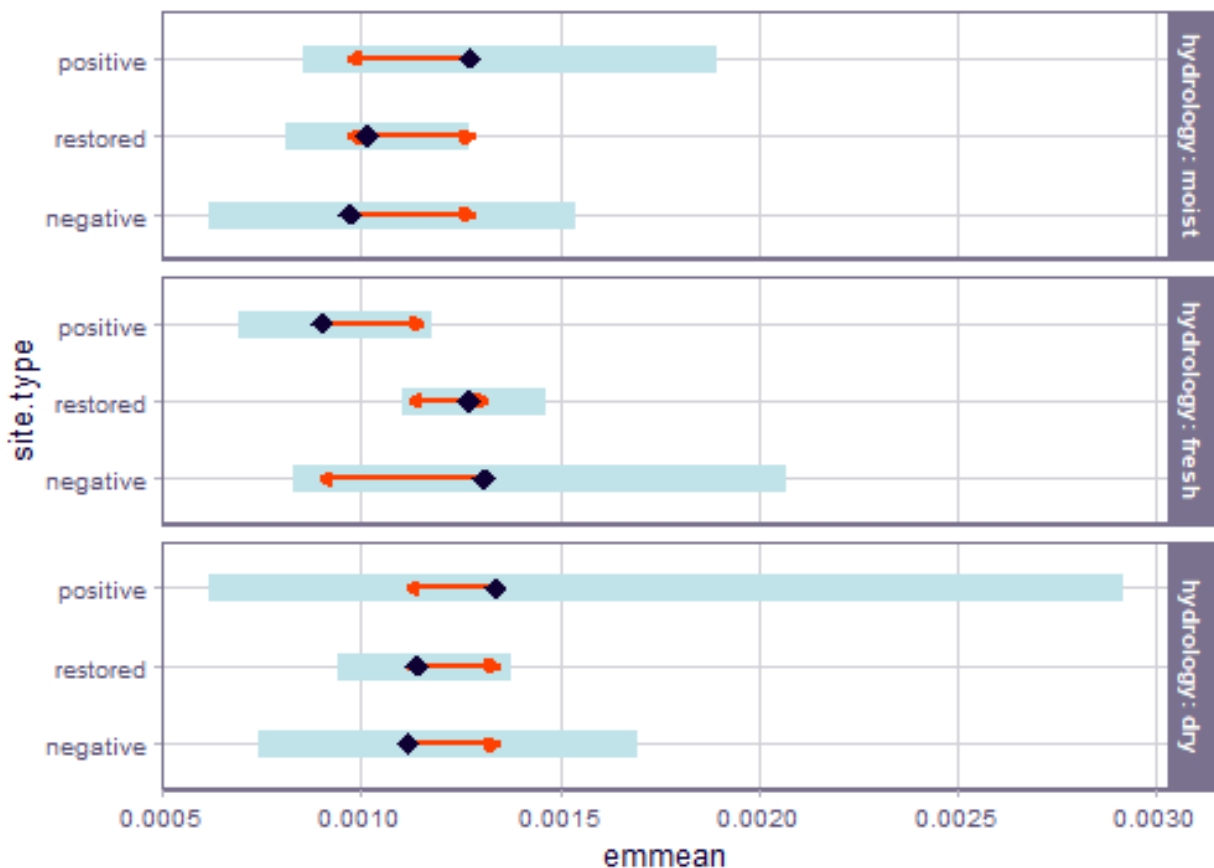
```
emm
```

```
## $emmeans
##   site.type response      SE    df lower.CL upper.CL
##   negative  0.00113 1.40e-04  14.1 0.000862  0.00147
##   restored  0.00114 6.25e-05 177.5 0.001021  0.00127
##   positive  0.00115 1.78e-04  52.5 0.000848  0.00157
##
## Results are averaged over the levels of: hydrology, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
##   contrast      ratio    SE    df null t.ratio p.value
##   restored / negative  1.01 0.138 16.7    1   0.081  0.9964
##   positive / negative  1.03 0.203 21.6    1   0.131  0.9906
```

```
## positive / restored 1.01 0.166 61.4    1    0.091  0.9955
##
## Results are averaged over the levels of: hydrology, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
```

Effect sizes: Hydrology x site type

```
emm <- emmeans(
  m_1,
  revpairwise ~ site.type | hydrology,
  type = "response"
)
plot(emm, comparison = TRUE)
```



```
emm
```

```
## $emmeans
## hydrology = moist:
## site.type response      SE      df lower.CL upper.CL
## negative  0.000974 2.27e-04 454.41 0.000616 0.00154
## restored  0.001016 1.15e-04 177.38 0.000812 0.00127
## positive  0.001275 2.55e-04  98.36 0.000857 0.00190
##
## hydrology = fresh:
```

```

## site.type response      SE      df lower.CL upper.CL
## negative  0.001308 3.07e-04 2760.94 0.000826 0.00207
## restored  0.001272 9.03e-05 177.50 0.001105 0.00146
## positive  0.000902 1.14e-04 15.48 0.000690 0.00118
##
## hydrology = dry:
## site.type response      SE      df lower.CL upper.CL
## negative  0.001119 2.03e-04 8.29 0.000739 0.00169
## restored  0.001140 1.10e-04 177.44 0.000942 0.00138
## positive  0.001339 5.32e-04 3941.22 0.000614 0.00292
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## hydrology = moist:
## contrast      ratio      SE      df null t.ratio p.value
## restored / negative 1.044 0.270 342.28 1 0.165 0.9851
## positive / negative 1.309 0.401 161.58 1 0.879 0.6546
## positive / restored 1.254 0.289 109.52 1 0.984 0.5886
##
## hydrology = fresh:
## contrast      ratio      SE      df null t.ratio p.value
## restored / negative 0.972 0.238 952.34 1 -0.116 0.9926
## positive / negative 0.690 0.182 24.66 1 -1.407 0.3530
## positive / restored 0.710 0.103 17.47 1 -2.363 0.0730
##
## hydrology = dry:
## contrast      ratio      SE      df null t.ratio p.value
## restored / negative 1.019 0.209 9.06 1 0.092 0.9953
## positive / negative 1.197 0.523 11.78 1 0.411 0.9118
## positive / restored 1.174 0.480 1190.57 1 0.393 0.9184
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale

```

Session info

```

## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
## LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8 LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8

```

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##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_2.0.3   DHARMA_0.5.0   patchwork_1.3.2 ggbeeswarm_0.7.3
## [5] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
## [9] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2     tibble_3.3.1
## [13] ggplot2_4.0.3   tidyverse_2.0.0 here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.6      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.2.0       magrittr_2.0.5     multcomp_1.4-30
## [7] ote1_0.2.0        compiler_4.5.2     mgcv_1.9-3
## [10] vctr_0.7.3        pkgconfig_2.0.3    crayon_1.5.3
## [13] fastmap_1.2.0     backports_1.5.1    labeling_0.4.3
## [16] utf8_1.2.6        ggstance_0.3.7     promises_1.5.0
## [19] rmarkdown_2.31    tzdb_0.5.0         nloptr_2.2.1
## [22] bit_4.6.0         xfun_0.58          later_1.4.8
## [25] broom_1.0.13      parallel_4.5.2     R6_2.6.1
## [28] gap.datasets_0.0.6 stringi_1.8.7       qgam_2.0.0
## [31] RColorBrewer_1.1-3 car_3.1-5           boot_1.3-32
## [34] estimability_1.5.1 Rcpp_1.1.1-1.1     iterators_1.0.14
## [37] knitr_1.51        zoo_1.8-15         parameters_0.29.1
## [40] httpuv_1.6.17     Matrix_1.7-4       splines_4.5.2
## [43] timechange_0.4.0  tidyselct_1.2.1    rstudioapi_0.18.0
## [46] abind_1.4-8       yaml_2.3.12        MuMIn_1.48.19
## [49] doParallel_1.0.17 codetools_0.2-20    lattice_0.22-7
## [52] plyr_1.8.9        bayestestR_0.18.1  shiny_1.13.0
## [55] withr_3.0.2       S7_0.2.2           evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3     pillar_1.11.1
## [61] gap_1.15.2        carData_3.0-6      foreach_1.5.2
## [64] stats4_4.5.2      insight_1.5.1      reformulas_0.4.4
## [67] generics_0.1.4    vroom_1.7.1        rprojroot_2.1.1
## [70] hms_1.1.4         scales_1.4.0       minqa_1.2.8
## [73] xtable_1.8-8      glue_1.8.1         tools_4.5.2
## [76] data.table_1.18.4 lme4_2.0-1         mvtnorm_1.4-0
## [79] grid_4.5.2        rbibutils_2.4.1    datawizard_1.3.1
## [82] nlme_3.1-168      Rmisc_1.5.1        performance_0.17.0
## [85] beeswarm_0.4.0    vipor_0.4.7        Formula_1.2-5
## [88] cli_3.6.6         gtable_0.3.6       digest_0.6.39
## [91] pbkrtest_0.5.5    TH.data_1.1-5      farver_2.1.2
## [94] htmltools_0.5.9   lifecycle_1.0.5    mime_0.13
## [97] bit64_4.8.2       dotwhisker_0.8.4   MASS_7.3-65

```